MIL-A-52174B
21 March 1972
SUPERSEDING
MIL-A-52174A
5 July 1962

MILITARY SPECIFICATION

ALUMINUM ALLOY DUCT SHEET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aluminum alloy duct sheet.

1.2 Classification. The duct sheet shall be supplied in flat or coiled sheet form, in one temper, and in the following nominal thicknesses:

0.016 inch
0.018 inch
0.020 inch
0.022 inch
0.025 inch
0.032 inch
0.040 inch
0.050 inch
0.063 inch

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals; Test Methods
Fed. Std. No. 245 - Tolerances for Aluminum Alloy and Magnesium Alloy Wrought Products

FSC 9535

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MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-649 - Aluminum and Magnesium Products, Preparation for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 8 - Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition shall conform to the requirements shown in table I.

Table I. Chemical composition[1/]

Element	Percent, maximum
Silicon	1.0
Zinc	1.0
Iron	1.0
Magnesium	0.7
Manganese	1.5
Copper	0.40
Chromium	0.20
Titanium	0.20
Other elements, each	0.05
Other elements, total	0.15
Aluminum	Remainder

1/ Analysis shall regularly be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

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3.2 Mechanical properties.

3.2.1 Tensile strength of material as supplied. The tensile strength parallel to the direction of final rolling shall be not less than 16,000 psi.

3.2.2 Forming properties. The sheet shall be capable of being formed into a Pittsburgh Lock Seam (see 4.3.2.2 and 6.3).

3.3 Dimensional variations. Sheet shall not vary from the specified dimensions by amounts greater than the applicable tolerance specified for squareness of sheet, thickness of 3003 aluminum alloy flat and coiled sheet; width of sheared flat sheet, width of sheared coiled sheet, and lateral bow of sheet of Fed. Std. No. 245. The length of flat sheet shall not vary more than plus or minus 1/4 inch from the specified length.

3.4 Marking for identification. Each sheet and coil shall be marked in at least one location with the words "Aluminum Duct Sheet" or equivalent terminology.

3.5 Workmanship. All sheet shall be uniform in quality and condition, and shall be clean, sound, smooth, commercially flat and free from buckles and other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of all sheet of the same thickness and temper submitted for inspection at one time.

4.2.2 Sampling for chemical analysis.

4.2.2.1 Ingot analysis. At least one sample shall be taken from each group of ingots of the same alloy poured simultaneously from the same source of molten metal by the producer and analyzed to determine conformance to 3.1. Ingots not conforming to the requirements of this specification shall be rejected. Complete ingot analysis records shall be available at the producer's plant to the Government.

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4.2.2.2 Finish product analysis. When compliance with 4.2.2.1 cannot be established, samples shall be selected as follows: One sample shall be taken for each 2000 pounds or less of sheet comprising a lot, except that not more than one sample shall be taken from the same sheet, when more than one piece is available.

4.2.3 Samples and specimens for mechanical tests.

4.2.3.1 Number of tests. One tension test shall be made for each 2000 pounds or less of sheet comprising the lot, except that in no case shall less than two tension tests represent a lot. Not more than one tension test specimen shall be taken from the same sheet when a lot consists of more than one sheet.

4.2.3.2 Locations of test specimens. Tension test specimens shall be taken from the sheet parallel to the direction of final rolling.

4.3 Inspection.

4.3.1 Chemical analysis. Chemical analysis shall be made by wet chemical or spectrochemical methods in accordance with Fed. Test Method Std. No. 151, method 111 or 112, respectively. In case of dispute, the chemical analysis by the wet chemical method (method 111) shall be the basis for acceptance.

4.3.2 Mechanical property tests.

4.3.2.1 Tension test. The tension test shall be conducted in accordance with ASTM E 8.

4.3.2.1.1 Types of test specimens. Tension test specimens shall conform to the requirements of ASTM E 8. A standard rectangular tension test specimen of sheet type, 1/2 inch wide (figure 6) of ASTM E 8 shall be used. Sheet less than 3/4 inch in width shall be tested in full section.

4.3.2.2 Bend test. Specimens shall be capable of being formed into a Pittsburgh Lock Seam either parallel with or transversely to the direction of final rolling, at room temperature (see 3.2.2 and 6.3).

4.3.3 Examination.

4.3.3.1 Dimensional and workmanship. Each lot shall be examined in accordance with the requirements of this specification for dimensional variations and workmanship.

4.3.3.2 Preparation for shipment. Examination of the packaging, packing, and marking for shipment shall be made for conformance to the requirements of section 5.

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4.4 Rejection and retests.

4.4.1 Rejection. Material failing to meet any of the requirements of this specification shall be rejected.

4.4.2 Retests. Retests will be permitted in accordance with the provisions of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. All sheet shall be preserved, packaged, and packed in accordance with the requirements of MIL-STD-649. The procuring activity should specify the levels required. Unless otherwise specified (see 6.2), material shall be preserved, packaged, and packed in accordance with level C.

5.2 Marking. In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Aluminum alloy duct sheet should be used only for duct work applications in heating, ventilating, and air conditioning systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Thickness desired (see 1.2).

c. Level of packaging and packing desired (see section 5).

d. Width desired (see 6.4).

6.3 Pittsburgh Lock Seam. A Pittsburgh Lock Seam is as shown in figure 1.

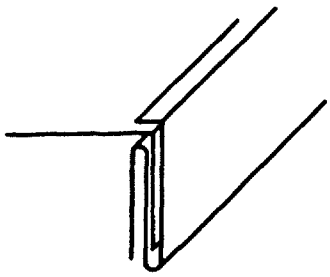


Figure 1. - Pittsburgh Lock seam

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6.4 The standard widths of aluminum alloy duct sheet are 24, 30, 36, and 48 inches.

Custodians:
Army - MR
Air Force - 84

Review activities:
Army - Mr, MU
DSA - IS

User activities:
Army - GL, AT
Navy - YD
Air Force - 11

Preparing activity:
Army - MR

Project No. 9535-0183

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-A-52174B, Aluminum Alloy Duct Sheet

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM, APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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